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| **Course Name:** Systems & Signals Processing | **Date:** May 19, 2008 |
| **Course No:** 402-322 | **ExamTime:** 50 min |
| **Instructor:** Eng. Juma Abu-Ghalune | **No. of pages:** 1 |

[redacted]

***Answer only 4 problems from the 5 problems below.***

1. Find the energy spectral density of the signal $x(t) = 8sinc(10t)$ then find the total energy of this signal.

2. Consider the signal $x(t) = 2 + 6.\cos^2(150\pi t) + \sin(100\pi t)\cos(200\pi t)$. Find:
   a. The fundamental period.
   b. The Fourier series coefficients.
   c. The average power.

3. Consider a fixed linear system with amplitude response $3.\Pi[w/600\pi] + 3.\Pi[w/400\pi]$ and phase response $-0.002\,w$ for $|w| \leq 250\pi$, $0.5\pi$ for $w < -250\pi$, and $-0.5\pi$ for $w > 250\pi$. Plot the amplitude and the phase of the transfer function of the system then find the output and determine the type of distortion for the following inputs:
   a. $x_1(t) = 2.\cos(225\pi t) + 4.\cos(275\pi t)$.
   b. $x_2(t) = 4.\cos(150\pi t) + 2.\cos(225\pi t)$.

4. Consider a periodic signal $x(t)$ is defined in the fundamental period $-2 < t < 2$ by $x(t) = 2t$, for $|t| < 1$, and $x(t) = 0$, for $1 < |t| < 2$. Find:
   a. The Complex Fourier series of $x(t)$.
   b. $X_n$, $a_n$, and $b_n$ for $n < 5$.
   c. The average power contained in the first four harmonics.

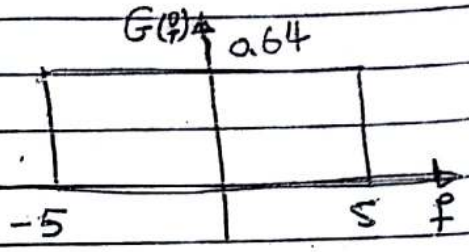
5. Using the integral definition form find the :
   a. Fourier transform of $x(t) = 2.\exp(0.2t).\delta(t - 0.5)$.
   b. Inverse Fourier transform of $Y(f) = 4.\exp(0.5f).u(2 - f)$.

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Q.1 $x(t) = 8\,\text{sinc}(10t) = 0.8\,(10\,\text{sinc}(10t))$

$X(f) = 0.8\,\Pi\left(\frac{f}{10}\right)$.

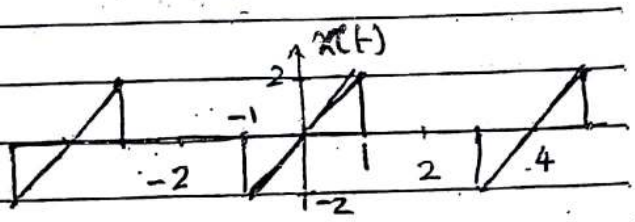
$G(f) = |X(f)|^2 = 0.64\,\Pi\left(\frac{f}{10}\right)$




$E = 2\int_0^{\infty} G(f)\,df = 2(0.64)\int_0^{5} df$

$= 6.4$ J.

Q.4 $x(t) = \begin{cases} 2t & -1 < t < 1 \\ 0 & 1 < |t| < 2 \end{cases}$




a $X_n = \frac{1}{4}\int_{-1}^{1} 2t\, e^{-j2\pi \frac{n}{2} t}\,dt = \frac{2}{2}\int_0^1 t\, e^{-j2\pi f_0 t n}\,dt$

$= \frac{t}{-j2\pi n f_0} e^{-j2\pi n f_0 t}\Big]_0^1 - \frac{1}{(j2\pi n f_0)^2} e^{-j2\pi n f_0 t}\Big]_0^1$

$= j\frac{2}{n\pi}\left[e^{-j2\pi n f_0} - \right] + \frac{4}{n^2\pi^2}\left[e^{-j2\pi n f_0} - 1\right]$

$= -\frac{4}{n^2\pi^2} + \frac{4}{n^2\pi^2} e^{-j\frac{n\pi}{4}} + j\frac{2}{n\pi} e^{-j\frac{n\pi}{4}}$

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Q.5 a. $x(t) = 2e^{0.2t}\,\delta(t-0.5)$.

$$X(f) = 2\int_{-\infty}^{\infty} e^{(0.2+j2\pi f)t}\,\delta(t-0.5)\,dt$$

$$= 2e^{0.5(0.2-j2\pi f)} = 2e^{(0.1+j\pi f)}$$

b. $Y(f) = 4e^{0.5f}\,u(2-f)$

$$y(t) = 4\int_{-\infty}^{2} e^{(0.5+j2\pi t)f}\,df = \frac{4}{[0.5+j2\pi t]}\,e^{2(0.5+j2\pi t)}$$

$$= \frac{4}{[0.5+j2\pi t]}\cdot e^{(1+j4\pi t)}$$

Q.2 $x(t) = 2 + 6\cos^2(150\pi t) + \sin(100\pi t)\cos(200\pi t)$

$= 5 - 0.5\sin(100\pi t) + 3\cos(300\pi t) + 0.5\sin(300\pi t)$.

a. $T_0 = \frac{2\pi}{100\pi} = 0.02\,s$

b. $X_0 = 5$; $X_1 = +j\frac{1}{4}$, $X_3 = 1.5 - j0.25$

c. $P_{avg} = 25 + 2\left[(0.25)^2 + (1.5)^2 + (0.25)^2\right]$

$= 29.75\,W$.

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